(i)

$$\mathrm{g}(x) = \ln(x+1) - \ln x - \frac{x+c}{x(x+1)}$$

$$\frac{d}{dx}\Big(\ln(x+1) - \ln x\Big) = \frac{1}{x+1} - \frac{1}{x} = -\frac{1}{x(x+1)}$$

$$\begin{aligned}\frac{d}{dx}\left(\frac{x+c}{x^2+x}\right) &= \frac{(x^2+x)-(x+c)(2x+1)}{(x^2+x)^2} \\ &= \frac{x^2+x-(2x^2+x+2cx+c)}{x^2(x+1)^2} \\ &= -\frac{x^2+2cx+c}{x^2(x+1)^2}\end{aligned}$$

$$\begin{aligned}\mathrm{g}'(x) &= -\frac{1}{x(x+1)} + \frac{x^2+2cx+c}{x^2(x+1)^2} \\ &= \frac{-x(x+1)+x^2+2cx+c}{x^2(x+1)^2} \\ &= \frac{(2c-1)x+c}{x^2(x+1)^2}\end{aligned}$$

$$x > 0 \implies x^2(x+1)^2 > 0$$

(i)(a) For $c \geq \dfrac{1}{2}$

$$2c-1 \geq 0, \quad c > 0 \implies (2c-1)x + c > 0$$

$$\therefore \boxed{\mathrm{g}'(x) > 0 \text{ for all } x > 0}$$

(i)(b) For $0 \leq c < \dfrac{1}{2}$

$$(2c-1)x + c < 0 \iff x > \frac{c}{1-2c}$$

$$\boxed{x > \frac{c}{1-2c}}$$

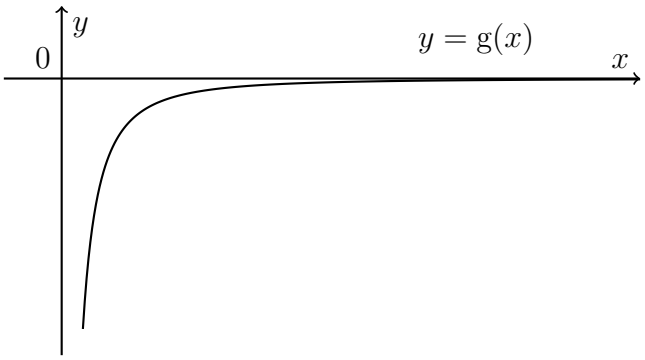
(ii)

As $x \to \infty$, $\ln\left(1+\dfrac{1}{x}\right) \to 0$ and $\dfrac{x+c}{x(x+1)} \to 0 \implies \mathrm{g}(x) \to 0$

(ii)(a) $c = \dfrac{3}{4}$

$\because$ g is increasing on $x > 0$

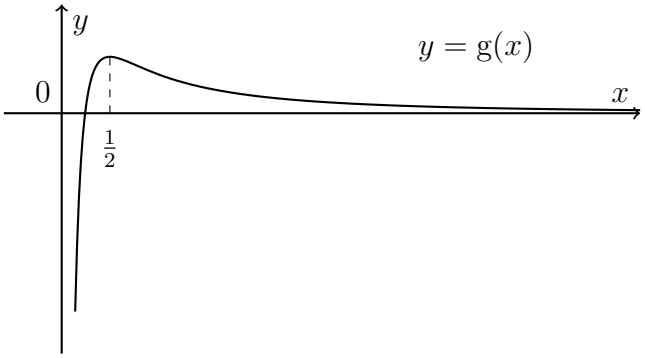
$\therefore \mathrm{g}(x) < 0$



(ii)(b) $c = \dfrac{1}{4}$

$$\frac{c}{1-2c} = \frac{1}{2} \implies \text{maximum point at } x = \frac{1}{2}$$

$$\mathrm{g}\left(\frac{1}{2}\right) = \ln 3 - 1 > 0$$



(iii)

$$\ln \mathrm{f}(x) = (x+c)\ln\left(1+\frac{1}{x}\right)$$

$$\begin{aligned}\text{Let } \mathrm{g}(x) = \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} &= \ln\left(1+\frac{1}{x}\right) + (x+c)\left(-\frac{1}{x(x+1)}\right) \\ &= \ln\left(1+\frac{1}{x}\right) - \frac{x+c}{x(x+1)}\end{aligned}$$

(iii)(a) For $c \geq \dfrac{1}{2}$, from part (ii), $\mathrm{g}(x) < 0$ for all $x > 0$

$$\implies \mathrm{f}'(x) < 0$$

$$\therefore \boxed{\text{f is decreasing}}$$

(iii)(b) For $0 < c < \dfrac{1}{2}$, let $a = \dfrac{c}{1-2c}$

g is continuous and increasing on $(0, a)$, with $\mathrm{g}(x) \to -\infty$ as $x \to 0$ and $\mathrm{g}(a) > 0$, so g has exactly one root $x_0 \in (0, a)$.

$$\mathrm{g}(x) < 0 \text{ on } (0, x_0), \qquad \mathrm{g}(x) > 0 \text{ on } (x_0, \infty)$$

$$\implies \mathrm{f}'(x) < 0 \text{ on } (0, x_0), \qquad \mathrm{f}'(x) > 0 \text{ on } (x_0, \infty)$$

$$\therefore \boxed{\text{f has a minimum point at } x = x_0}$$

(iii)(c) For $c = 0$

$$\mathrm{g}'(x) = \frac{-x}{x^2(x+1)^2} < 0$$

g is decreasing on $x > 0$ with $\mathrm{g}(x) \to 0$, so $\mathrm{g}(x) > 0$ throughout.

$$\implies \mathrm{f}'(x) > 0$$

$$\therefore \boxed{\text{f is increasing}}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: September 1, 2026